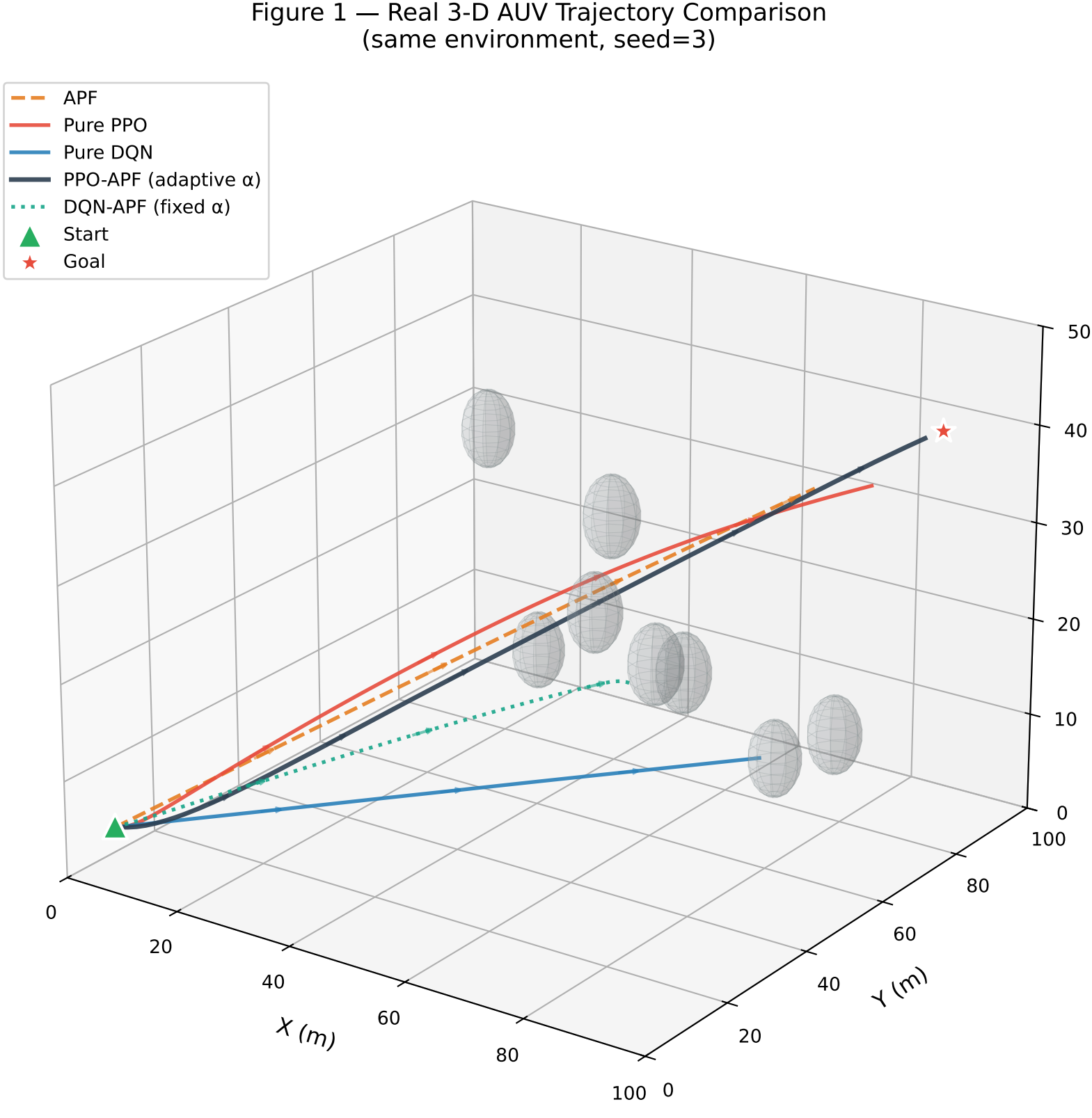

Figure 1 — Real 3-D AUV Trajectory Comparison
(same environment, seed=3)
APF
Pure PPO
Pure DQN
PPO-APF (adaptive α)
DQN-APF (fixed α)
Start
Goal
0
20
40
60
80
100
X (m)
0
20
40
60
80
100
Y (m)
0
10
20
30
40
50